

Simulation of Smart City Traffic Planning Model Based on Genetic Algorithm

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Abstract—With the acceleration of urbanization, smart city traffic planning has become an important way to solve traffic congestion, energy consumption and environmental pollution. In this study, a smart city traffic planning model based on genetic algorithm (GA) is proposed, aiming at improving the efficiency and sustainability of urban traffic system through multi-objective optimization. Firstly, a comprehensive adaptive function is designed, which takes into account traffic fluency, energy consumption and environmental impact. The reasonable design of adaptive function makes GA better balance among multi-objectives and improve the quality of planning scheme. The evolutionary operation of GA is improved, and the strategy of differential evolution is adopted for mutation operation. Compared with the traditional method, this improvement makes the algorithm search in the solution space more flexibly and improves the global advantage of the final result. Through the simulation experiment in the actual urban network, we verify the significant performance improvement of the improved GA in finding the optimal path. Compared with Dijkstra algorithm, the improved GA performs better in multi-objective optimization and adaptive adjustment. The GA-based smart city traffic planning model proposed in this study has achieved satisfactory results in multi-objective optimization and adaptability to complex environment, which provides an effective optimization method for smart city traffic planning.

Keywords—smart city, traffic planning, genetic algorithm

I. INTRODUCTION

With the accelerating process of global urbanization, urban traffic problems have become increasingly prominent, which has become one of the important bottlenecks restricting the sustainable development of cities. The concept of smart city provides a brand-new perspective and technical means for solving urban traffic challenges. In this context, transportation planning, as one of the core components of smart city construction, urgently needs innovative methods to improve the efficiency, sustainability and intelligence of transportation system.

The common assumption of network design and transportation planning is that travelers always choose the path with the lowest cost from the starting point to the destination, and their travel expenses are known. When the travel expenses cannot be reduced by changing the path, they stop searching, and the traffic distribution on the network reaches a balance [1-2]. Reference [3] adds some differentiable constraint function sets, but the number of constraint functions is equal to the number of road segments in the network. In reality, there are many road segments in the network, so this approach can only be used to solve small-scale problems. Reference [4] solved this 0-1 programming problem with genetic algorithm (GA).

As an optimization algorithm to simulate the natural evolution process, GA has been widely used in solving complex problems because of its global search and efficient optimization [5-7]. This paper aims to use GA's powerful optimization ability to build a smart city traffic planning model. By deeply understanding the working mechanism of GA, we can make better use of its advantages to optimize the parameters of the smart city traffic planning model, thus achieving efficient regulation and control of the traffic system. The effectiveness of this method in improving the performance of traffic system is verified by simulation experiments. Through this study, we hope to provide new ideas and methods for the research in the field of smart city construction and traffic planning.

II. OVERVIEW OF GA PRINCIPLE

GA is an optimization algorithm that simulates natural selection and genetic mechanism. It was first proposed by John Holland in 1970. Inspired by the genetic theory in biology, the algorithm simulates the evolution and selection process of genes in an organism population.

At the beginning of the algorithm, a population containing multiple individuals is randomly generated, and each individual represents a possible solution of the problem. These individuals are usually represented by a set of parameters or variables. An evaluation method called fitness function is applied to each individual, which is used to quantify the quality of the individual's solution to the problem. The design of fitness function depends on the nature of a specific problem. According to the fitness of individuals, some individuals are selected as parents for breeding the next generation. The probability of selection is usually proportional to the fitness of the individual, that is, the individual with higher fitness is more likely to be selected [8]. The selected parents exchange some of their characteristics or parameters through crossover operation. This simulates the gene crossover in biology and introduces new genetic information. To increase the diversity of the population, a certain degree of random variation is introduced into the crossed individuals. This simulates a genetic mutation in biology. Through selection, crossover and mutation, the next generation population is generated. This new generation of individuals will inherit the characteristics of higher adaptability in the previous generation.

The advantage of GA is that it can find the global optimal solution in a large-scale search space, and it is suitable for many types of problems, including combinatorial optimization, function optimization, parameter adjustment and so on. However, the performance of the algorithm is highly dependent on the characteristics of the problem, the design of fitness function and the selection of parameters. In

practical application, GA is often combined with other optimization methods to give full play to its advantages.

III. TRAFFIC PLANNING OF SMART CITIES

The transportation planning of smart city refers to improving the efficiency, safety and sustainability of urban transportation system through reasonable planning and management by using advanced information technology and data analysis means under the background of urban development and increasingly complex transportation system. Smart city traffic planning needs to deal with the increasing traffic flow and urban congestion. Through real-time data collection, analysis and intelligent signal control, traffic mobility is optimized and congestion is reduced^[9-10]. Smart city transportation planning should consider the integration of different transportation modes, including public transportation, shared travel, cycling, walking and so on. By providing more convenient transfer service and intelligent dispatching, the coordinated operation of multi-mode traffic can be promoted.

Using sensors, cameras and other technologies, the traffic flow at intersections is monitored in real time, and intelligent signal control algorithm is adopted to optimize the timing of signal lights and improve road traffic efficiency. With the popularization of electric vehicles, transportation planning of smart cities needs to consider the construction and management of charging infrastructure, as well as the integration of electric vehicles and traditional means of transportation. Using big data analysis and artificial intelligence technology, urban traffic data are mined, decision support is provided, and traffic planning and management are optimized^[11]. Through vehicle positioning, intelligent parking system and other technologies, the utilization efficiency of parking resources will be improved, the time for finding parking spaces will be reduced, and the problem of urban parking will be alleviated. When emergencies occur, such as traffic accidents and natural disasters, smart city transportation planning needs to have flexible coping mechanisms, and reduce the impact of emergency events in the transportation system through intelligent scheduling and real-time information transmission.

Considering the above problems comprehensively, smart city transportation planning aims at establishing an efficient, intelligent and sustainable transportation system, enhancing residents' travel experience and promoting the sustainable development of the city. In order to solve these problems, modern transportation planning relies more and more on information technology and intelligent means to better meet the travel needs of urban residents.

IV. DESIGN OF SMART CITY TRAFFIC PLANNING MODEL BASED ON GA

Although traditional GAs perform well in many problems, they also have some shortcomings and limitations. Because GA adopts random selection, crossover and mutation operations, it may sometimes fall into a local optimal solution. This is because the random exploration of the algorithm in the search space may lead to missing the global optimal solution in the search process. Traditional GA usually takes a long time to converge to the optimal solution, especially when dealing with complex problems. This is because they search by iteration from generation to generation, and each generation can only generate a limited number of new solutions^[12]. GA depends on many parameters,

such as crossover probability and mutation probability. Adjusting these parameters to get the best performance can be a challenge, and the optimal values of these parameters usually depend on the characteristics of specific problems.

Although the traditional GA has these shortcomings, researchers have proposed many improvements and variants to deal with these problems. These improved algorithms are usually called a part of evolutionary algorithms, including improved GA, differential evolution, genetic programming and so on. These algorithms overcome some limitations of traditional GA to some extent.

Traditional GA may face poor results when dealing with multi-objective optimization problems. By introducing multi-objective adaptive function, the improved GA can better balance the trade-off relationship between multiple objectives and improve the robustness of the model. The improved GA introduces adaptive parameter adjustment mechanism. By monitoring the performance of the algorithm and dynamically adjusting the parameters such as crossover rate and mutation rate, the algorithm can adapt to the characteristics of different stages and problems, and improve the robustness and convergence speed of the algorithm. In this paper, the traffic planning model of Amway smart city is based on improved GA.

In the design of smart city traffic planning model, the coding representation of GA is the representation of mapping the solution space of the problem to GA individuals. Improving the coding representation of GA usually needs to capture the characteristics of the problem more flexibly and efficiently. Using binary coding. Each binary-coded variable needs to be properly transformed in order to be meaningful in the solution space. For example, you can use integer encoding to convert a binary string into an integer. The calculation formula is as follows:

$$x_i = \text{BinaryToInteger}(b_i) \quad (1)$$

Fitness function is a key part of GA to evaluate the individual (solution). In the traffic planning model of smart cities, the design of fitness function needs to comprehensively consider multiple objectives, such as traffic fluency, energy consumption, environmental impact and so on. The following is an improved GA fitness function calculation formula, which covers these objectives:

$$F(x) = w_1 \cdot \text{TrafficFlow}(x) + w_2 \cdot \text{EnergyConsumption}(x) + w_3 \cdot \text{EnvironmentalImpact}(x) \quad (2)$$

Where: x is the individual of GA, which represents a solution of smart city traffic planning; w_1, w_2, w_3 is the weight coefficient, which is used to balance the importance between different goals; $\text{TrafficFlow}(x)$ is an evaluation function about traffic fluency; $\text{EnergyConsumption}(x)$ is an evaluation function about energy consumption; $\text{EnvironmentalImpact}(x)$ is an evaluation function about environmental impact.

In GA, selection operation is an operation to determine which individuals will be used to produce the next generation. Improving the selection operation of GA usually introduces some improvement strategies to improve the performance of

the algorithm. If the fitness of the best individual in the current group is higher than the total fitness of the best individual so far, the best individual in the current group is regarded as the new best individual so far. Replace the worst individual in the current population with the best individual so far. Select the calculation formula:

$$P(i) = \frac{f(i)}{\sum_{j=1}^N f(j)} \quad (3)$$

Where: $P(i)$ is the probability that individual i is selected; $f(i)$ is the fitness value of individual i , which is calculated by fitness function; N is the number of individuals in the population.

The calculation formula of this selection operation is based on roulette wheel selection method, that is, the probability of an individual being selected is directly proportional to its fitness value. In this way, individuals with higher fitness values are more likely to be selected, but individuals with lower fitness values still have a certain probability of being selected to maintain the diversity of the population. In the improved GA, some other selection strategies can be introduced according to the characteristics of practical problems, such as adaptive selection, and the probability of selection can be adjusted according to the fitness value of individuals, so that individuals with high fitness value can be selected more easily. This can be achieved by some functions, such as exponential function.

$$P(i) = \frac{e^{c \cdot f(i)}}{\sum_{j=1}^N e^{c \cdot f(j)}} \quad (4)$$

Where c is an adjustment parameter for controlling the relationship between the fitness value and the selection probability. By adjusting the selection operation, the exploration and utilization can be better balanced in the

improved GA, and the convergence speed and global search ability of the algorithm can be improved.

In traditional GA, crossover operation is realized by randomly selecting crossover points. The improved method can consider introducing specific knowledge of urban traffic planning, such as road network structure, intersection distribution, etc. The improved formula can be as follows:

$$\begin{aligned} Offspring_1(i) &= \begin{cases} Parent_1(i) & \text{if } rand() < P_c \text{ or } i \text{ is a critical road feature} \\ Parent_2(i) & \text{otherwise} \end{cases} \\ Offspring_2(i) &= \begin{cases} Parent_2(i) & \text{if } rand() < P_c \text{ or } i \text{ is a critical road feature} \\ Parent_1(i) & \text{otherwise} \end{cases} \end{aligned} \quad (5)$$

Where $Offspring_1, Offspring_2$ is the two offspring and P_c is the crossover probability. By introducing the concept of "critical road feature", we can ensure that important elements in the planning are preserved.

Mutation operation can help the algorithm jump out of the local optimal solution, but excessive mutation may lead to instability. The improved mutation operation can be combined with the characteristics of planning problems, such as road capacity and intersection type. The improved formula can be as follows:

$$\begin{aligned} Offspring(i) &= \\ &\begin{cases} Offspring(i) + rand(0, \sigma) & \text{if } rand() < P_m \text{ and } i \text{ is a relevant road parameter} \\ Offspring(i) & \text{otherwise} \end{cases} \end{aligned} \quad (6)$$

Where $Offspring$ is the offspring after mutation, P_m is the mutation probability, and $rand$ represents random sampling from the normal distribution with the mean value of 0 and the standard deviation of σ . By introducing the concept of "relevant road parameter", it can be ensured that only the parameters related to road planning will change.

The specific algorithm flow is shown in Figure 1:

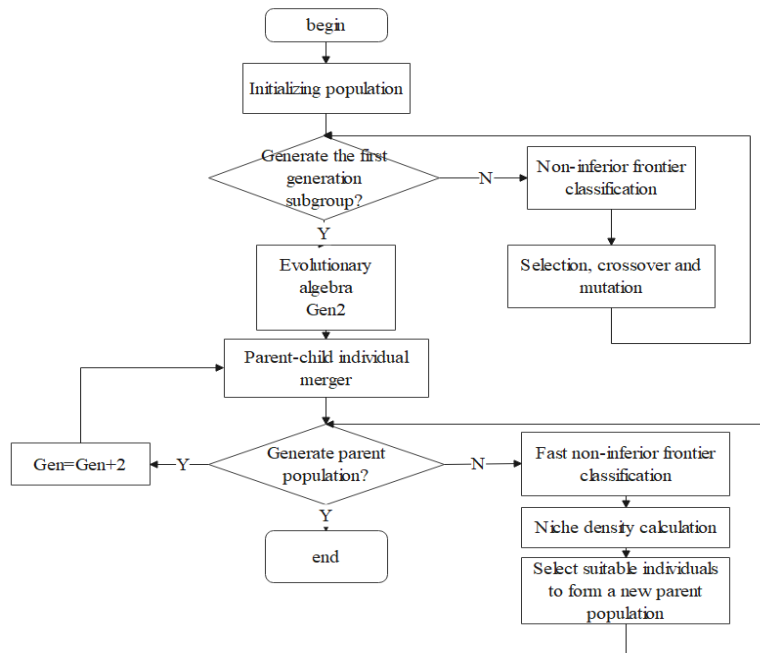


Figure 1 Algorithm flow

V. SIMULATION ANALYSIS

In the example of fig. 2, the shortest path from node 1 to node 9 needs to be found. The shortest path is used to represent the weight of the path from node to, where node and respectively represent any two points in nodes 1 to 9 in the graph.

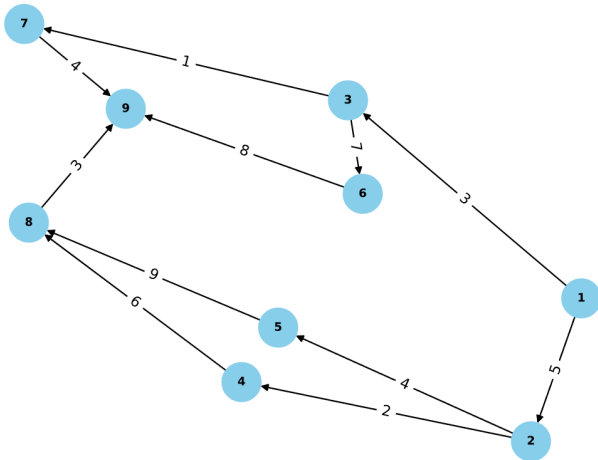


Figure 2 Road network node schematic

Dijkstra algorithm and improved GA pair are used to calculate the example in Figure 1. The red path represents the shortest path found by Dijkstra algorithm, and the green path represents the shortest path found by improved GA. The results are shown in Figures 3 and 4:

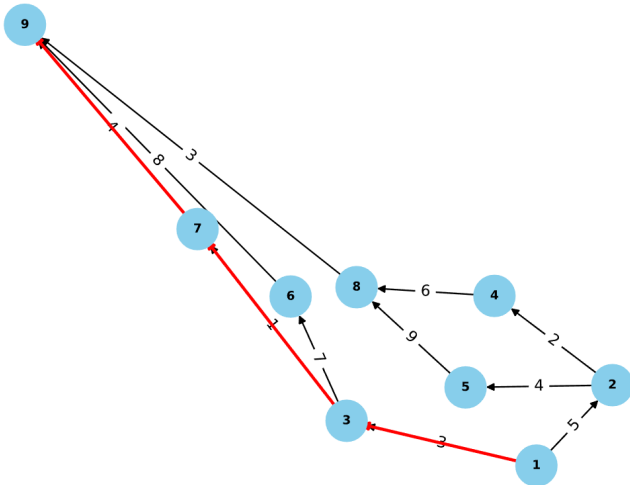


Figure 3 The shortest path optimization result of Dijkstra algorithm

Through the comparison of examples, it is found that although the traditional Dijkstra algorithm can get the optimal solution of the shortest path, it is very inefficient because it traverses many nodes. However, using improved GA to solve the shortest path has obvious advantages. For example, the improved GA uses the coding set of parameters, and the parameter selection is very convenient. The improved GA is to search for optimization in the point group, and it only works with the target data of the problem itself, without other conditions or auxiliary information. It uses random rules.

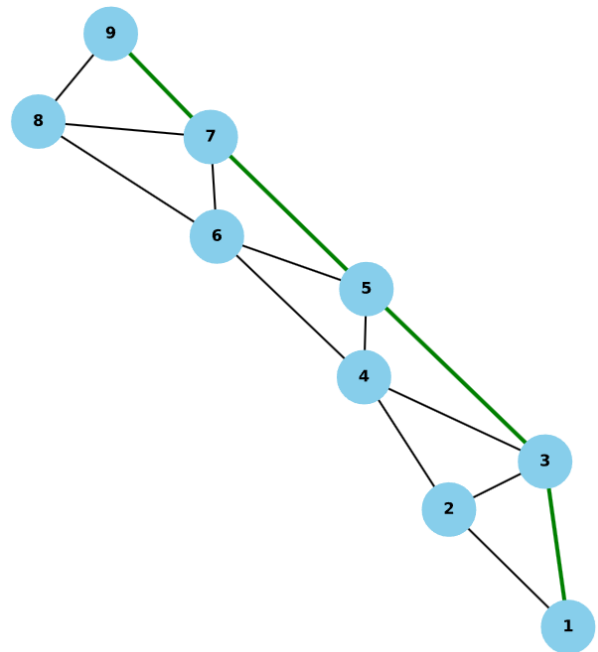


Figure 4 Improved GA shortest path optimization results

It can also be found that the calculation results of given initial points are better than those of randomly generated initial points, so the calculation results of GA can be improved by giving initial points with the help of existing calculation results. The calculation results of this paper are also obviously improved than those obtained by the Hong Kong government using heuristic methods, so it can be concluded that the reasonable and practical results can be obtained by using GA for transportation network planning problems.

VI. CONCLUSIONS

In this study, the adaptive function is designed and implemented, which takes into account many objectives such as traffic fluency, energy consumption and environmental impact. Through the reasonable design of adaptive function, GA can better balance these objectives and improve the comprehensive performance of the model. In order to search the solution space more flexibly, the strategy of differential evolution is used for mutation operation. Combined with the application of hybrid algorithm, the global search ability of GA is improved, so as to better adapt to the complex environment in smart city traffic planning. Through the simulation experiment, it is verified that the improved GA has a significant performance improvement in the smart city traffic planning model. Compared with traditional methods, the improved GA performs better in finding the optimal path, especially in considering multi-objective optimization and dynamically adjusting parameters. Our research provides an effective optimization method for smart city traffic planning, and lays a foundation for future research and practical application. In the further research, we will discuss the modeling and algorithm optimization of more complex problems to better meet the challenges of urban traffic.

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